CLEM with AMR: energy write-back failure at the injector rim

CrossJet case 1 (H_2 jet in Mach 1.5 crossflow), non-reacting, branch particle-clem. Diagnosis of 2026-10-14.

1. Summary

With two refinement levels the run stops with *CLEM splice: inflow from a LES ghost cell with no state*. The splice is only where the problem is detected. The cause is the energy channel of the conserved-delta write-back: in the first wall cell at the rim of the injector, the line energy that overwrites the LES energy is lower, every step, than what the LES energy equation gives. The cell cools and expands without bound (1000 K to 163 K, 1 atm to 0.1 atm) until the LES returns NaN.

The same coupling, unchanged from CLEMNew, survives on the grid CLEMNew was validated on (single level, 0.3125 mm) and with one refinement level. It fails only when the injector lip is resolved by the finest level (0.078 mm). Mass is conserved to 10^{-15} throughout, so the splice, the reflux, the halo and the average-down are not involved.

2. The coupling as implemented

Identical in CLEMNew (`ClemLesCoupling.cpp, writeBackProjected`, `clem.couple_back_energy = 1`) and in the AMR code (`coupling/write_back_coupling.cpp`). The LES advances ρ, ρ**u**, ρE with the species frozen, giving U^{**}. The CLEM step then writes back:

$$\Delta[\rho e] = \Sigma_p\, m_p\, e_p / V_c \text{ (after the CLEM operator)} - \Sigma_p\, m_p\, e_p / V_c \text{ (before it)}$$

$$(\rho e)^{n+1} = (\rho E)^n - k^n + \Delta[\rho e]\,, \quad (\rho E)^{n+1} = (\rho e)^{n+1} + k^{n+1}$$

The bracket covers the whole CLEM operator: isentropic pressure update to the LES pressure, diffusion, splice, regrid and volume renormalisation. **$(\rho E)^{**}$, the total energy the LES equation solved, does not appear: it is overwritten.** It is used only indirectly, through the LES pressure the lines are driven to. The energy of a cell is therefore what the lines carry: transport by the splice plus isentropic p dV work. Terms present in the LES energy equation but not on the lines are dropped every step: viscous heating (including at the no-slip wall), heat conduction between cells, and the dissipation of shocks and unresolved expansions.

3. Where and how it fails

Two levels, impulsive jet (`prob.jet_ramp = 0`). Failure at coarse step 78, t = 6.84×10^{-7} s, level-2 substep 308. Cell (544, 0, 32) of level 2: x = 4.254 cm, z = 0.254 cm, first cell off the lower wall, 0.54 mm from the jet axis, i.e. just outside the 0.5 mm hole radius. The ghost cell at j = –1 is its wall reflection. A per-cell probe (`Source/Clem/probe.cpp`) recorded its history:

Level-2 substep	ρ [g/cm^3]	Y_{H2}	T [K]	p [dyn/cm^2]
0	3.51e-4	0	1000	1.01e6
~225	3.24e-5	0.96	199	2.56e5
307	1.57e-5	0.9996	163	1.05e5
308	NaN	NaN	NaN	NaN

There is no sudden event: T falls by about 0.5 K and p by about 1.4 % on every substep. The neighbour over the hole, (543, 0, 32), sits at p ≈ 1.0×10^6 and T ≈ 280 K, so the pressure ratio between the two adjacent cells is about 10. Over the run the level-2 write-back *max volume defect* rose from 0.4 % to

17.5 %.

4. The energy budget of one step

Specific internal energy of cell (544, 0, 32) through substep 307, the last finite one:

Stage	Line e [erg/g]	Line volume / V_c
Start of the step	–2.6006e10	1.000
After the pressure update (to the LES pressure 1.10e5)	–2.6318e10	0.943
After diffusion	–2.6318e10	0.943
After splice (arrivals at T 280 K, p 1.0e6)	–2.4875e10	0.842
After volume renormalisation = written back	–2.6055e10	1.000
LES energy equation, (ρE^{**} – k)/ρ	**–2.5817e10**	–

For the same step the LES energy equation raises e by 1.9×10^8 erg/g; the lines lower it by 0.5×10^8. The write-back keeps the line value, so 2.4×10^8 erg/g (about 2.4 K) is removed from the cell every substep. Lower energy gives a lower LES pressure, which makes the cell expand more on the next step, which the lines again treat isentropically: a positive feedback.

Not yet measured: how the missing 2.4×10^8 erg/g splits between wall friction heating, inter-cell conduction and the isentropic treatment of the factor-10 expansion. The frozen-species LES state ($\Sigma\rho Y_k / \rho = 1.012$ in this cell) biases the LES temperature by about 3 K and the target pressure by a few percent; it is a minor contribution, not the driver.

5. Evidence

All runs: half channel (6.25 × 1.5 × 0.5 cm, base 200 × 48 × 16) or the full channel where noted, 64 MPI ranks, same executable unless noted.

Run	Outcome	Injector-rim cell
2 levels, CLEM compiled out (LES only, full channel)	ran to step 110, no failure	–
2 levels, impulsive jet	**NaN at 0.68 µs**	163 K and falling
2 levels, impulsive jet, energy write-back off (LES keeps ρE^{**})	ran to step 110 (1.03 µs)	minimum 193 K, recovers to 241 K
0 levels (CLEMNew resolution), impulsive jet	ran 600 steps (7.1 µs)	settles near 185 K, still drifting slowly
1 level, impulsive jet	ran 300 steps (3.2 µs)	minimum 172 K, recovers to 230 K
2 levels, jet ramped over 10 µs (as in CLEMNew)	**NaN at 12.0 µs**, step 972	same failure at rim cells (541, 0, 37) and (541, 0, 26)

Reading of the table:

- The LES alone and the LES keeping its own energy both survive: the failure needs the energy overwrite.
- The coupling survives at 0 and 1 level and fails at 2: the error grows with resolution. On the 0.3125 mm grid the 1 mm injector spans about 3 cells and the lip expansion is smeared; at 0.078 mm the rim cell sees the full jump from the cell over the hole.
- The 10 µs ramp used in CLEMNew only delays the failure, to 2 µs after the jet reaches full speed.

CLEMNew ran this case only with `amr.max_level = 0`, a 10 µs jet ramp and stirring on; none of its runs reached the resolution at which the coupling breaks.

6. Separate bug found and fixed on the way

The ramped run first stopped at step 886 with *ParticleContainer::locateParticle(): invalid particle*, in `clem::AdaptToGrids` during a regrid that removed part of level 1. The code assumed AMReX drops the lines of cells that leave a level; AMReX aborts instead. Fix (`multiresolution/grid_adapter.cpp`): before refiling, the elements whose cell is no longer on the level are marked invalid, so the redistribute removes them. Tested with a refinement box that expires: level 2 lost 1.5 million cells (18,087,936 elements) and level 1 went from 319,488 to 106,496 cells; the line mass matched the LES to 4×10^{-15} on every level afterwards. The restarted ramped run passed three such regrids before the energy failure.

7. Options for the energy channel

- **A (recommended).** Keep $(\rho E)^{**}$ from the LES as the total energy and return from the lines only what the LES does not contain (diffusion along the line now; reaction and stirring later), by taking the $\Delta[\rho e]$ bracket around those stages only. Transport and p dV work stay with the LES, which handles them conservatively. Its limiting case (no line energy at all) is the variant shown to survive at two levels.
- **B.** Keep the current formula and add the LES terms the lines lack (viscous and conduction parts of the LES energy source) to $(\rho e)^{n} + \Delta[\rho e]$. Needs those terms split out of the MOL source, and does not address the isentropic treatment of large pressure jumps.

Either changes the coupling validated in CLEMNew, so it is a modelling decision. Until it is made, the one-level cost run (`inputs-case1-clem-amr1.inp`) can proceed; the two-level run cannot reach 0.6 ms.

8. Code changes made during the diagnosis

- `Source/Clem/probe.cpp` (new): per-cell probe of the LES state and the line at each stage of the CLEM step; off unless `clem.probe_level` and `clem.probe_cells` are given.
- `setup.H`: `write_back_energy` switch, default true (behaviour unchanged).
- `splicing/splicer.cpp`: the ghost-cell check prints the cell and its state before aborting.
- `multiresolution/grid_adapter.cpp`: the regrid fix of section 6.